Estimated angle in the local coordinate

2026-10-28 local coordinate of the estimate (u, v, w); measurement coordinate (x, y, z)

1 Gain matrix in the local coordinate of the estimate

$$
\begin{aligned}
{}^{V[k]}\hat{\mathbf{A}}[k] &= \hat{\mathbf{V}}^T[k]\,\hat{\mathbf{A}}[k]\,\hat{\mathbf{V}}[k] \approx \operatorname{diag}\big(\hat{a}_{11}[k],\ \hat{a}_{22}[k],\ \hat{a}_{33}[k]\big) \\
{}^{V[k]}\bar{\mathbf{A}}[k+1] &= \hat{\mathbf{V}}^T[k]\,\bar{\mathbf{A}}[k+1]\,\hat{\mathbf{V}}[k] \\
&\approx \hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1]\,\operatorname{diag}\big(\hat{a}_{11}[k+1],\ \hat{a}_{22}[k+1],\ \hat{a}_{33}[k+1]\big)\ \hat{\mathbf{V}}^T[k+1]\,\hat{\mathbf{V}}[k] \\
\hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1] &= \mathbf{R}_x^T(\hat{\theta}_x[k])\,\mathbf{R}_y^T(\hat{\theta}_y[k])\,\mathbf{R}_y\big(\hat{\theta}_y[k]+\delta\hat{\theta}_y[k]\big)\,\mathbf{R}_x\big(\hat{\theta}_x[k]+\delta\hat{\theta}_x[k]\big) \\
&= \mathbf{R}_x^T(\hat{\theta}_x[k])\,\mathbf{R}_y(\delta\hat{\theta}_y[k])\,\mathbf{R}_x(\hat{\theta}_x[k])\,\mathbf{R}_x(\delta\hat{\theta}_x[k])
\end{aligned}
$$

small angle

$$
\mathbf{R}_y(\delta\hat{\theta}_y) \approx \begin{bmatrix} 1 & 0 & \delta\hat{\theta}_y \\ 0 & 1 & 0 \\ -\delta\hat{\theta}_y & 0 & 1 \end{bmatrix}, \qquad \mathbf{R}_x(\delta\hat{\theta}_x) \approx \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\hat{\theta}_x \\ 0 & \delta\hat{\theta}_x & 1 \end{bmatrix}
$$

product of two small angles left out

$$
\hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1] \approx \begin{bmatrix} 1 & \sin\hat{\theta}_x[k]\,\delta\hat{\theta}_y[k] & \cos\hat{\theta}_x[k]\,\delta\hat{\theta}_y[k] \\ -\sin\hat{\theta}_x[k]\,\delta\hat{\theta}_y[k] & 1 & -\delta\hat{\theta}_x[k] \\ -\cos\hat{\theta}_x[k]\,\delta\hat{\theta}_y[k] & \delta\hat{\theta}_x[k] & 1 \end{bmatrix}
$$

$$
{}^{V[k]}\bar{\mathbf{A}}[k+1] \approx \begin{bmatrix} \hat{a}_{11}[k+1] & 0 & -\big(\hat{a}_{11}[k]-\hat{a}_{33}[k]\big)\cos\hat{\theta}_x[k]\,\delta\hat{\theta}_y[k] \\ 0 & \hat{a}_{22}[k+1] & \big(\hat{a}_{22}[k]-\hat{a}_{33}[k]\big)\,\delta\hat{\theta}_x[k] \\ -\big(\hat{a}_{11}[k]-\hat{a}_{33}[k]\big) & \big(\hat{a}_{22}[k]-\hat{a}_{33}[k]\big) & \hat{a}_{33}[k+1] \\ \times\cos\hat{\theta}_x[k]\,\delta\hat{\theta}_y[k] & \times\delta\hat{\theta}_x[k] & \end{bmatrix}
$$

$$
= \begin{bmatrix} \hat{a}_{11}[k+1] & 0 & \hat{a}_{13}[k+1] \\ 0 & \hat{a}_{22}[k+1] & \hat{a}_{23}[k+1] \\ \hat{a}_{13}[k+1] & \hat{a}_{23}[k+1] & \hat{a}_{33}[k+1] \end{bmatrix}
$$

2 Process in the local coordinate

left superscript v: local coordinate of $\hat{\mathbf{V}}$; as Section 1 with $\theta - \hat{\theta}$ in place of $\delta\hat{\theta}$

$$
{}^{v}\bar{\mathbf{A}} = \hat{\mathbf{V}}^{T}\,\bar{\mathbf{A}}\,\hat{\mathbf{V}} = \begin{bmatrix} \bar{a}_{11} & 0 & \bar{a}_{13} \\ 0 & \bar{a}_{11} & \bar{a}_{23} \\ \bar{a}_{13} & \bar{a}_{23} & \bar{a}_{33} \end{bmatrix}
$$

$$
\bar{a}_{23} = \big(\bar{a}_{11} - \bar{a}_{33}\big)\big(\theta_x - \hat{\theta}_x\big)
$$

$$
\bar{a}_{13} = -\big(\bar{a}_{11} - \bar{a}_{33}\big)\cos\hat{\theta}_x\,\big(\theta_y - \hat{\theta}_y\big)
$$

motion control law

$$
{}^{v}\bar{\mathbf{f}}_d[k] = {}^{v}\hat{\mathbf{A}}^{-1}[k]\left\{{}^{v}\bar{\mathbf{p}}_d[k+1] - {}^{v}\hat{\mathbf{p}}[k] - \lambda_c\left({}^{v}\bar{\mathbf{p}}_d[k] - {}^{v}\hat{\mathbf{p}}[k]\right)\right\}
$$

dynamics of tracking error; F_u, F_v, F_w: elements of ${}^{v}\bar{\mathbf{F}}_{dp}[k]$

$$
{}^{v}\delta\bar{\mathbf{p}}[k+1] = \lambda_c\,{}^{v}\delta\bar{\mathbf{p}}[k] - \left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}[k]\right){}^{v}\bar{\mathbf{F}}_{dp}[k] - {}^{v}\boldsymbol{\varepsilon}_p[k]
$$

$$
\left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}\right){}^{v}\bar{\mathbf{F}}_{dp} = \mathbf{G}_a[k]\begin{bmatrix} \bar{a}_{11} - \hat{a}_{11} & \bar{a}_{33} - \hat{a}_{33} & \bar{a}_{13} - \hat{a}_{13} & \bar{a}_{23} - \hat{a}_{23} \end{bmatrix}^{T}
$$

$$
\mathbf{G}_a[k] = \begin{bmatrix} F_u & 0 & F_w & 0 \\ F_v & 0 & 0 & F_w \\ 0 & F_w & F_u & F_v \end{bmatrix}
$$

motion along the surface normal; ${}^{v}\mathbf{w}$: true normal in the local coordinate of the estimate (third column of $\hat{\mathbf{V}}^{T}\mathbf{V}$, small-angle form of the matrix above with $\theta - \hat{\theta}$)

$$
{}^{v}\mathbf{M}[k] = {}^{v}\Delta\bar{\mathbf{p}}_d[k] + (1-\lambda_c)\,{}^{v}\delta\bar{\mathbf{p}}_3[k]
$$

$$
{}^{v}\mathbf{w} = \begin{bmatrix} \cos\hat{\theta}_x\,(\theta_y - \hat{\theta}_y) & -(\theta_x - \hat{\theta}_x) & 1 \end{bmatrix}^{T}
$$

true state

$$
{}^{v}\delta\bar{\mathbf{p}}_1[k+1] = {}^{v}\delta\bar{\mathbf{p}}_2[k]
$$

$$
{}^{v}\delta\bar{\mathbf{p}}_2[k+1] = {}^{v}\delta\bar{\mathbf{p}}_3[k]
$$

$$
{}^{v}\delta\bar{\mathbf{p}}_3[k+1] = \lambda_c\,{}^{v}\delta\bar{\mathbf{p}}_3[k] - \left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}[k]\right){}^{v}\bar{\mathbf{F}}_{dp}[k] - {}^{v}\boldsymbol{\varepsilon}_p[k]
$$

$$
\bar{a}_{11}[k+1] = \bar{a}_{11}[k] + \bar{a}'_{11}[k]\;{}^{v}\mathbf{w}^{T}\left({}^{v}\mathbf{M}[k] + \left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}[k]\right){}^{v}\bar{\mathbf{F}}_{dp}[k] + {}^{v}\boldsymbol{\varepsilon}_p[k]\right)
$$

$$
\bar{a}_{33}[k+1] = \bar{a}_{33}[k] + \bar{a}'_{33}[k]\;{}^{v}\mathbf{w}^{T}\left({}^{v}\mathbf{M}[k] + \left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}[k]\right){}^{v}\bar{\mathbf{F}}_{dp}[k] + {}^{v}\boldsymbol{\varepsilon}_p[k]\right)
$$

$$
\bar{a}'_{11}[k+1] = \bar{a}'_{11}[k]
$$

$$
\bar{a}'_{33}[k+1] = \bar{a}'_{33}[k]
$$

$$
\bar{a}_{13}[k+1] = \bar{a}_{13}[k]
$$

$$
\bar{a}_{23}[k+1] = \bar{a}_{23}[k]
$$

$$
\boldsymbol{\theta}[k+1] = \boldsymbol{\theta}[k]
$$

3 Estimator

$$\begin{bmatrix} {}^v\delta\bar{\mathbf{p}}_1^T & {}^v\delta\bar{\mathbf{p}}_2^T & {}^v\delta\bar{\mathbf{p}}_3^T & \bar{a}_{11} & \bar{a}_{33} & \bar{a}'_{11} & \bar{a}'_{33} & \bar{a}_{13} & \bar{a}_{23} & \theta_x & \theta_y \end{bmatrix}^T \qquad (17\times 1,\ d=2)$$

process outputs in the local coordinate

$${}^v\delta\bar{\mathbf{p}}_m[k] = \hat{\mathbf{V}}^T[k]\,\delta\bar{\mathbf{p}}_m[k], \qquad {}^v\bar{\mathbf{A}}_m[k] = \hat{\mathbf{V}}^T[k]\,\bar{\mathbf{A}}_m[k]\,\hat{\mathbf{V}}[k]$$

$$\bar{a}_{m,11} = \tfrac{1}{2}\Big(\big[{}^v\bar{\mathbf{A}}_m\big]_{11} + \big[{}^v\bar{\mathbf{A}}_m\big]_{22}\Big), \qquad \bar{a}_{m,33} = \big[{}^v\bar{\mathbf{A}}_m\big]_{33}, \qquad \bar{a}_{m,13} = \big[{}^v\bar{\mathbf{A}}_m\big]_{13}, \qquad \bar{a}_{m,23} = \big[{}^v\bar{\mathbf{A}}_m\big]_{23}$$

process outputs, their estimates and the output matrix (no measurement of the angles)

$$\mathbf{y}_m[k] = \begin{bmatrix} {}^v\delta\bar{\mathbf{p}}_m[k] \\ \bar{a}_{m,11}[k] \\ \bar{a}_{m,33}[k] \\ \bar{a}_{m,13}[k] \\ \bar{a}_{m,23}[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} {}^v\delta\hat{\mathbf{p}}_1[k] \\ \hat{a}_{11}[k] \\ \hat{a}_{33}[k] \\ \hat{a}_{13}[k] \\ \hat{a}_{23}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_2 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_2 & \mathbf{0} \end{bmatrix}$$

feedback: measured − estimate ($\mathbf{e}_a$: the four gain elements)

$$\mathbf{e}_p[k] = {}^v\delta\bar{\mathbf{p}}_m[k] - {}^v\delta\hat{\mathbf{p}}_1[k]$$

$$\mathbf{e}_a[k] = \begin{bmatrix} \bar{a}_{m,11} - \hat{a}_{11} & \bar{a}_{m,33} - \hat{a}_{33} & \bar{a}_{m,13} - \hat{a}_{13} & \bar{a}_{m,23} - \hat{a}_{23} \end{bmatrix}^T$$

estimator

$${}^v\hat{\mathbf{M}}[k] = {}^v\Delta\bar{\mathbf{p}}_d[k] + (1-\lambda_c)\,{}^v\delta\hat{\mathbf{p}}_3[k]$$

$${}^v\delta\hat{\mathbf{p}}_1[k+1] = {}^v\delta\hat{\mathbf{p}}_2[k] + \mathbf{L}_1[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$${}^v\delta\hat{\mathbf{p}}_2[k+1] = {}^v\delta\hat{\mathbf{p}}_3[k] + \mathbf{L}_2[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$${}^v\delta\hat{\mathbf{p}}_3[k+1] = \lambda_c\,{}^v\delta\hat{\mathbf{p}}_3[k] + \mathbf{L}_3[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$$\hat{a}_{11}[k+1] = \hat{a}_{11}[k] + \hat{a}'_{11}[k]\left[{}^v\hat{\mathbf{M}}[k]\right]_w + \mathbf{L}_4[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$$\hat{a}_{33}[k+1] = \hat{a}_{33}[k] + \hat{a}'_{33}[k]\left[{}^v\hat{\mathbf{M}}[k]\right]_w + \mathbf{L}_5[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$$\hat{a}'_{11}[k+1] = \hat{a}'_{11}[k] + \mathbf{L}_6[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$$\hat{a}'_{33}[k+1] = \hat{a}'_{33}[k] + \mathbf{L}_7[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$$\hat{a}_{13}[k+1] = \hat{a}_{13}[k] + \mathbf{L}_8[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$$\hat{a}_{23}[k+1] = \hat{a}_{23}[k] + \mathbf{L}_9[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T$$

$$\hat{\boldsymbol{\theta}}[k+1] = \hat{\boldsymbol{\theta}}[k] + \delta\hat{\theta}[k]$$

$$\delta\hat{\theta}_x[k] = \frac{\hat{a}_{23}[k+1]}{\hat{a}_{11}[k] - \hat{a}_{33}[k]}$$

$$\delta\hat{\theta}_y[k] = -\frac{\hat{a}_{13}[k+1]}{\big(\hat{a}_{11}[k] - \hat{a}_{33}[k]\big)\cos\hat{\theta}_x[k]}$$

local coordinate of $\hat{\mathbf{V}}[k+1]$; $\hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1]$ of Section 1

$${}^v\delta\hat{\mathbf{p}}_i \leftarrow \big(\hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1]\big)^T\,{}^v\delta\hat{\mathbf{p}}_i$$

$${}^{V[k+1]}\hat{\mathbf{A}}[k+1] \approx \mathrm{diag}\big(\hat{a}_{11}[k+1],\ \hat{a}_{22}[k+1],\ \hat{a}_{33}[k+1]\big), \qquad \hat{a}_{13}[k+1] \leftarrow 0, \qquad \hat{a}_{23}[k+1] \leftarrow 0$$

4 Dynamics of estimation errors

estimation error: true − estimate

$$\mathbf{e}[k] = \big[\, \big({}^{v}\delta\bar{\mathbf{p}}_1 - {}^{v}\delta\hat{\mathbf{p}}_1\big)^T \;\; \big({}^{v}\delta\bar{\mathbf{p}}_2 - {}^{v}\delta\hat{\mathbf{p}}_2\big)^T \;\; \big({}^{v}\delta\bar{\mathbf{p}}_3 - {}^{v}\delta\hat{\mathbf{p}}_3\big)^T$$
$$\bar{a}_{11} - \hat{a}_{11} \;\; \bar{a}_{33} - \hat{a}_{33} \;\; \bar{a}'_{11} - \hat{a}'_{11} \;\; \bar{a}'_{33} - \hat{a}'_{33}$$
$$\bar{a}_{13} - \hat{a}_{13} \;\; \bar{a}_{23} - \hat{a}_{23} \;\; \big(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\big)^T \,\big]^T$$

angle: first order; $\bar{a}_{23}[k+1]$, $\bar{a}_{13}[k+1]$ in the local coordinate of $\hat{\mathbf{V}}[k]$: $\bar{a}_{23} = \big(\bar{a}_{11} - \bar{a}_{33}\big)(\theta_x - \hat{\theta}_x[k])$, $\bar{a}_{13} = -\big(\bar{a}_{11} - \bar{a}_{33}\big)\cos\hat{\theta}_x(\theta_y - \hat{\theta}_y[k])$

$$\theta_x - \hat{\theta}_x[k+1] = \big(\theta_x - \hat{\theta}_x[k]\big) - \frac{\hat{a}_{23}[k+1]}{\hat{a}_{11}[k] - \hat{a}_{33}[k]} = \frac{\bar{a}_{23}[k+1] - \hat{a}_{23}[k+1]}{\hat{a}_{11}[k] - \hat{a}_{33}[k]}$$
$$\theta_y - \hat{\theta}_y[k+1] = \big(\theta_y - \hat{\theta}_y[k]\big) + \frac{\hat{a}_{13}[k+1]}{\big(\hat{a}_{11}[k] - \hat{a}_{33}[k]\big)\cos\hat{\theta}_x[k]} = -\frac{\bar{a}_{13}[k+1] - \hat{a}_{13}[k+1]}{\big(\hat{a}_{11}[k] - \hat{a}_{33}[k]\big)\cos\hat{\theta}_x[k]}$$

other states: first order, feedback terms left out; all elements at step k

$${}^{v}\delta\bar{\mathbf{p}}_3[k+1] - {}^{v}\delta\hat{\mathbf{p}}_3[k+1] = \lambda_c\big({}^{v}\delta\bar{\mathbf{p}}_3 - {}^{v}\delta\hat{\mathbf{p}}_3\big) - \mathbf{G}_a\big[\,\bar{a}_{11} - \hat{a}_{11} \;\; \bar{a}_{33} - \hat{a}_{33} \;\; \bar{a}_{13} - \hat{a}_{13} \;\; \bar{a}_{23} - \hat{a}_{23}\,\big]^T - {}^{v}\boldsymbol{\varepsilon}_p$$
$$\begin{aligned}\bar{a}_{11}[k+1] - \hat{a}_{11}[k+1] &= \big(\bar{a}_{11} - \hat{a}_{11}\big) + \big(\bar{a}'_{11} - \hat{a}'_{11}\big)\big[{}^{v}\hat{\mathbf{M}}\big]_w + \hat{a}'_{11}(1-\lambda_c)\big[{}^{v}\delta\bar{\mathbf{p}}_3 - {}^{v}\delta\hat{\mathbf{p}}_3\big]_w \\ &\quad + \hat{a}'_{11}\Big[\mathbf{G}_a\big[\,\bar{a}_{11} - \hat{a}_{11} \;\; \bar{a}_{33} - \hat{a}_{33} \;\; \bar{a}_{13} - \hat{a}_{13} \;\; \bar{a}_{23} - \hat{a}_{23}\,\big]^T\Big]_w + \hat{a}'_{11}\big[{}^{v}\boldsymbol{\varepsilon}_p\big]_w \\ &\quad + \hat{a}'_{11}\Big(\cos\hat{\theta}_x\,\big[{}^{v}\hat{\mathbf{M}}\big]_u\,\big(\theta_y - \hat{\theta}_y\big) - \big[{}^{v}\hat{\mathbf{M}}\big]_v\,\big(\theta_x - \hat{\theta}_x\big)\Big)\end{aligned}$$
$$\begin{aligned}\bar{a}_{33}[k+1] - \hat{a}_{33}[k+1] &= \big(\bar{a}_{33} - \hat{a}_{33}\big) + \big(\bar{a}'_{33} - \hat{a}'_{33}\big)\big[{}^{v}\hat{\mathbf{M}}\big]_w + \hat{a}'_{33}(1-\lambda_c)\big[{}^{v}\delta\bar{\mathbf{p}}_3 - {}^{v}\delta\hat{\mathbf{p}}_3\big]_w \\ &\quad + \hat{a}'_{33}\Big[\mathbf{G}_a\big[\,\bar{a}_{11} - \hat{a}_{11} \;\; \bar{a}_{33} - \hat{a}_{33} \;\; \bar{a}_{13} - \hat{a}_{13} \;\; \bar{a}_{23} - \hat{a}_{23}\,\big]^T\Big]_w + \hat{a}'_{33}\big[{}^{v}\boldsymbol{\varepsilon}_p\big]_w \\ &\quad + \hat{a}'_{33}\Big(\cos\hat{\theta}_x\,\big[{}^{v}\hat{\mathbf{M}}\big]_u\,\big(\theta_y - \hat{\theta}_y\big) - \big[{}^{v}\hat{\mathbf{M}}\big]_v\,\big(\theta_x - \hat{\theta}_x\big)\Big)\end{aligned}$$
$$\bar{a}_{13}[k+1] - \hat{a}_{13}[k+1] = \bar{a}_{13} - \hat{a}_{13}$$
$$\bar{a}_{23}[k+1] - \hat{a}_{23}[k+1] = \bar{a}_{23} - \hat{a}_{23}$$

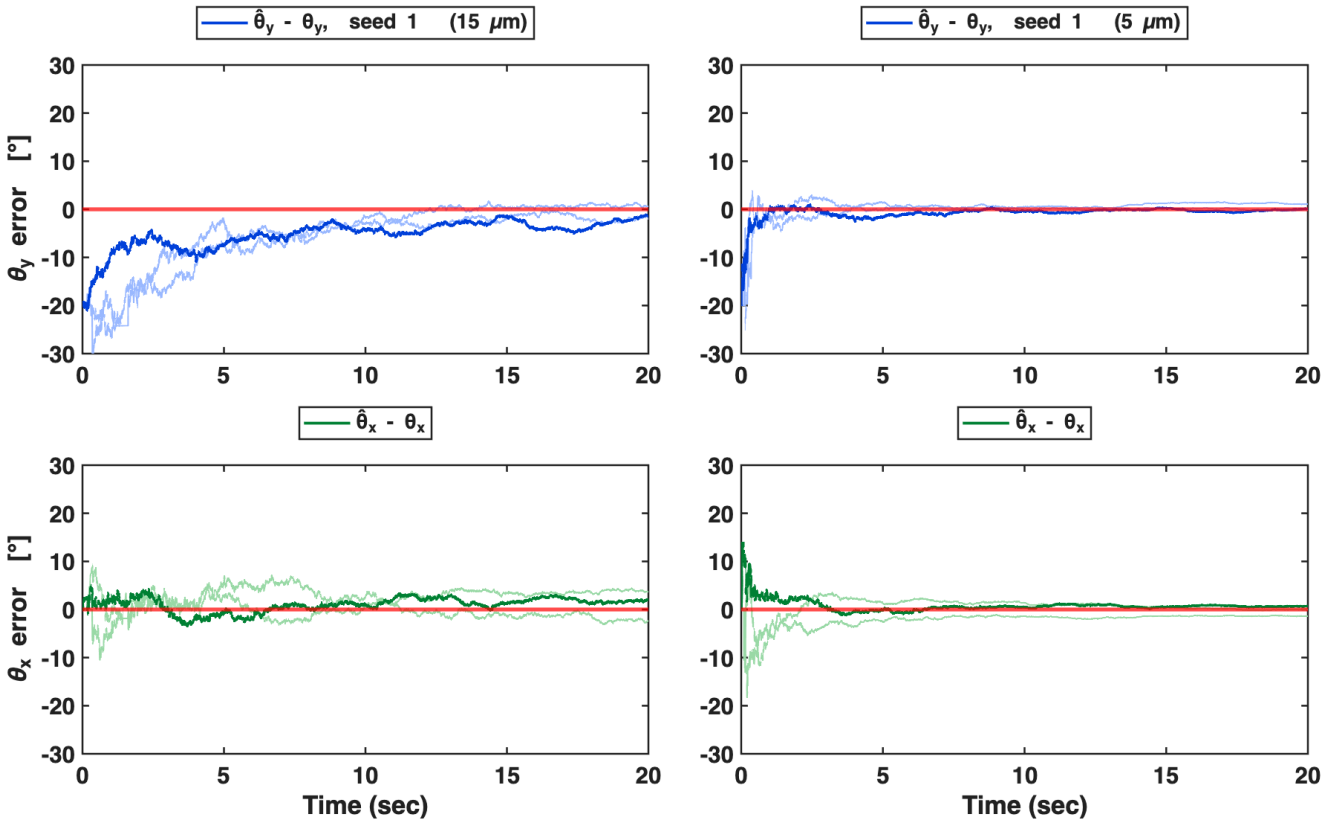
process noise

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] + \big[\,\mathbf{0} \;\; \mathbf{0} \;\; -{}^{v}\boldsymbol{\varepsilon}_p^T \;\; \hat{a}'_{11}[{}^{v}\boldsymbol{\varepsilon}_p]_w \;\; \hat{a}'_{33}[{}^{v}\boldsymbol{\varepsilon}_p]_w \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0\,\big]^T$$
$$\mathbf{Q}[k] = \mathrm{E}\big[\,(\cdot)(\cdot)^T\,\big] + \mathrm{diag}\big(\mathbf{0},\ \mathbf{0},\ \mathbf{0},\ 0,\ 0,\ \mathbf{Q}_{a'}[k],\ 0,\ 0,\ 0,\ 0\big)$$

estimator feedback matrix and error covariance: left open

5 Simulation at a fixed point

wall $\theta_y = 20°$, $\theta_x = 0$; probe held at 15 µm (left) and 5 µm (right); $\hat{\theta}[0] = \mathbf{0}$; 15 µm: 3 seeds with the smallest angle error of seeds 1–10, 5 µm: seeds 1–3; best seed dark



15 and 5 µm, seed 1; $\hat{\theta} = \hat{\theta}[0] + \sum \delta\hat{\theta}$

